CVPR #12319

MoKA: Multi-domain Knowledge Aggregation across Vision Foundation Models

Supplementary Material



(a) A driving video frame (b) A bird video frame

Figure 1. **Comparison on two video frames**. We visualize the RGB projection results of deep features utilizing PCA for Proteus [6], DINO-v3 [4], and our MoKA.

A. Demo

We include two video demos that compare our MoKA with the advanced Proteus [6] and DINO-v3 [4] methods. **We highly encourage the reviewers to take a glance at these demos** (`cars_comp.mp4` and `birds_comp.mp4` attached with this material) for a more intuitive understanding of the qualitative differences.

To create the demonstration video demos, we first download two short video clips from https://www.vecteezy.com/free-videos/. As illustrated in Fig. 1, one clip depicts a complex *driving* scenario, while the other contains a flock of *flying birds*. Due to the storage constraints of supplementary materials, each clip is trimmed to approximately five seconds and then decomposed into individual frames. For each individual frame, we extract deep features of a resolution of 128×128 (input image is resized to the size of 128 multiplying the patch size of ViT) using three pre-trained models: Proteus (ViT-B), DINO-v3 (ViT-B), and our MoKA (ViT-B). The resulting features are subsequently projected into the RGB space using PCA. Finally, the original frames and their corresponding PCA visualizations are combined into videos with 30 FPS for qualitative comparison.

Fig. 1 presents a qualitative comparison on a representative frame. Proteus exhibits pronounced grid artifacts caused by positional embedding interpolation. In contrast, DINO-v3 tends to miss small-scale objects, such as distant pedestrians and small birds. Our method, however, demonstrates more precise perception in both the driving and bird-flight scenarios, highlighting the effectiveness of **distilling multiple VFMs into a single versatile student**.

B. Implementation Details

In this section, we provide more details on the training of multi-teacher distillation and downstream benchmarks.

Multi-teacher distillation setting. We freeze all teachers during distillation, which forms a standard and default choice. Below, we provide details on the student and teachers:

- ResNet-50 student. We utilize the official ImageNet-1K classification pretrained, MAE pre-trained by SparK [5], and DINO pre-trained ResNet-50 models as teachers.
- ConvNeXt-B student. We adopt the official ImageNet-1K classification pre-trained, MAE pre-trained by SparK [5], and the latest DINO-v3 ConvNeXt-B models as teachers. We also adopt ViT teachers to explore distillation between hybrid architectures.
- ViT-B student. We implement a vanilla ViT-B/16 under the timm framework without register tokens and only using learnable position embedding. We employ the official CLIP ViT-B/16, MAE pre-trained ViT-B/16, along with the latest DINO-v3 ViT-B/16 as teachers. It is noteworthy that DINO-v3 employs advanced 2D ROPE position embeddings for better performance.
- ViT-L student. We implement a vanilla ViT-L/16 student using the official timm library. As for teachers, we adopt the official CLIP ViT-L/14, MAE pre-trained ViT-L/16, together with the latest DINO-v3 ViT-L/16 models.

Downstream benchmark setting. We conduct a performance evaluation of the distilled student on four fundamental vision tasks: ImageNet classification, fine-grained classification, image segmentation, and depth estimation.

We first fine-tune all listed models in Table 1 (main paper) except for Proteus and RADIO-v2.5 on the ImageNet-1K dataset for 10 epochs with a learning rate of 1e-4. The training starts with one epoch warmup. We set the batch size to 512. The training input resolution is 224×224. We adopt a layer-wise learning-rate decay of 0.8 for all models. Once tuning finished for the models, we report their top-1 accuracy on the validation set of ImageNet-1K.

For the dense segmentation on the ADE20K dataset, we adopt a linear decoder head following [2] while freezing the backbones, establishing a *linear probing* evaluation paradigm. Typically, we adopt a standard 80k training schedule with a training batch size of 16, utilizing the off-the-shelf MMSegmentation [1] toolkit. We report the mean Intersection over Union (mIoU) metric on the validation set. The training resolution is fixed as 512×512 for all models. To enable training on this resolution, we adopt a *bicubic position embedding interpolation* strategy.

Finally, we conduct downstream depth estimation on the widely adopted NYUv2 benchmark. Following [6], we use the DPT [3] as the decoder head. Specifically, the DPT decoder head requires hierarchical spatial features together with a summary token as inputs. Therefore, with ResNet-50 and ConvNeXt-B backbones, we utilize the hierarchical spatial features within four stages, and spatially pool these features to obtain a summary token. This design makes the DPT decoder head suitable for depth estimation with CNN models. As for the ViT-B backbone, we form the hierarchical deep features by extracting features at indices $3, 5, 7, 11$. Specifically, the corresponding `cls` tokens are regarded as the summary tokens required by the DPT decoder head. Additionally, the output indices for the ViT-L model are changed to $7, 11, 15, 23$. In practice, we train the student and teachers (except CLIP) on NYUv2 with a batch size of 16, experiencing a standard 25k training schedule. Following [6], the learning rate for the backbone is scaled by a factor of 0.1. Specifically, we adopt the Root Mean Square Error (RMSE) metric for evaluation.

C. Cross-space and Intra-space Module Design

CNN as student. We adopt two cross-space and intra-space modules for the CNN student. The cross-space module is comprised of layer normalization, followed by a depth convolution with 3×3 kernels, GELU activation, and a 1×1 convolution layer with G groups. We set G for ResNet-50 to 128 and 64 for ConvNeXt-B.

The intra-space module follows the same architecture design as the cross-space module. Additionally, the intra-space module is shared across different representational spaces. We adopt a residual connection to both types of modules.

Specifically, suppose the feature projected into M representational spaces (M means the number of teachers) has shape `(M, B, C, H, W)`. Notably, B is the batch size, C indicates the number of channels, and H, W represent the spatial resolution. We permute the feature into shape `(B, C, M, H, W)` and then reshape the feature into shape `(B, CM, H, W)` for the input of cross-modules. This enables a cross-space feature interaction among different projected spaces. For the input of intra-space modules, we reshape the feature into shape `(MB, C, H, W)`.

ViT as student. We build the cross-space and intra-space modules on top of the vanilla attention block, featuring a norm-attention-norm-ffn design. We adopt the widely utilized Layer Normalization as the implementation for normalization layers. Both types of modules share the same architecture. For the cross-space module, we enlarge the head number in the attention block linearly to the number of teachers. Additionally, the intra-space module is shared across multiple representational spaces.

Specifically, the feature shape is transformed in the same way for the CNN student above.

Table 1. **Alleviate the impact of dataset bias by choosing teachers** (Student as ConvNeXt-B).

Teacher-I	Teacher-II	Teacher-III	ADE20K	NYUv2 (RMSE)
CLIP (ViT-B/16)	MAE (ViT-B/16)	DINO-v3 (ViT-B/16)	34.7	0.349
DINO-v3 (ConvNeXt-B)	DINO-v3 (ViT-B/16)	-	35.1	0.332

D. Details on Alignment Losses

Model token alignment loss. After the alternating cross-space and intra-space modules, we chunk the model token from ultimate deep features at different representational spaces as illustrated in our main paper. We use a shared projection head (LayerNorm-Linear) to align its channels with all teachers. We then pool the model token to align with the teachers' global-level token. For CNN teachers, we utilize their gloabl average pooling (GAP) features as supervision. On the contrary, we adopt the `cls` token from ViT teachers except CLIP for supervision. For the CLIP teacher, we directly use its image encoding vector as the alignment target. We simply adopt MSE loss for all types of teachers.

Student feature alignment loss. After the alternating cross-space and intra-space modules, we chunk the student feature from the ultimate deep features. For classification-based and CLIP teachers, we pool the features and utilize a cosine loss for semantic alignment. For the MAE teacher, we adopt an MSE loss for patch-level alignment. As for the DINO teacher, we utilize a patch-level cosine constraint for the alignment, enabling semantic knowledge transferring.

E. Impact of Teachers

As illustrated in Sec. 4 (main paper), we can mitigate the influence of dataset bias among teachers and students by substituting all CNN teachers with ViTs. In this section, we observe that **reducing the dataset bias between teachers could also improve distillation performance**. As shown in Tab. 1, we adopt a DINO-v3 CNN and ViT teacher to distill a ConvNeXt-B student. Since the teachers are all pre-trained on the same LVD-1689M dataset, the dataset bias between them can be neglected. As a result, we obtain a better segmentation and depth estimation performance as shown in Tab. 1.

F. Igniting the Potential of Student

We continue our discussion about the multi-space feature interaction by analyzing the feature quality of a distilled ViT-B/16 student.

We visualize the PCA projected feature as well as the cosine similarity map between `cls` token and patch tokens. As shown in Figure 4 (main paper), the features from the backbone show satisfactory semantic information, while

the cosine map suggests that the `cls` token fails to capture discriminative information for classification. Furthermore, both the CLIP and MAE heads fail to simultaneously capture clean semantic clues together with discriminative information. In contrast, the DINO projected feature captures rich semantic information and presents the best concentration of the `cls` token. With these observations, we utilize the model token projected to the DINO head for classification. Analogously, we employ the deep features projected to the DINO head for dense prediction tasks.

References

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